

**67** Phosphatase enzymes break the bond between a phosphate group and the rest of a molecule.

Phenolphthalein phosphate (PPP) is colourless and is a substrate for phosphatase enzymes.

- The phosphate group of PPP can be removed by the phosphatase enzyme to produce the indicator phenolphthalein.
- Phenolphthalein goes pink in alkaline conditions.

Five tubes were set up with equal concentrations of PPP and a plant phosphatase enzyme.

Each tube was at a different pH. The tubes were incubated at the same temperature. After 10 minutes  $\text{Na}_2\text{CO}_3$  was then added to each tube until a pH of 9.5 was reached. The colour was observed immediately and then again after another 5 minutes.

| <i>tube</i>                                                                  | 1         | 2    | 3         | 4          | 5          |
|------------------------------------------------------------------------------|-----------|------|-----------|------------|------------|
| <i>pH of the initial reaction</i>                                            | 3.2       | 4.2  | 5.2       | 8.2        | 9.2        |
| <i>colour immediately after <math>\text{Na}_2\text{CO}_3</math> addition</i> | pale pink | pink | dark pink | colourless | colourless |
| <i>colour after another 5 minutes</i>                                        | pale pink | pink | dark pink | colourless | colourless |

Which of the following statements is/are correct?

- 1  $\text{Na}_2\text{CO}_3$  solution inhibited the activity of the phosphatase enzyme.
  - 2 More substrate was produced in conditions of high acidity compared to low acidity.
  - 3 The optimum pH for this plant phosphatase must be 5.2.
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3